Computer Science - K-12 Level Assessment 9 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. What is the value of this python expression: 1 + 3 % 3?
   (a) 0
   (b) 1
   (c) 3
   (d) 4
2. What is the output of "abc"[::-1] in Python 3?
   (a) Error
   (b) abc
   (c) cba
   (d) c
3. Which of the following spreadsheet functions would be most useful for detecting improbably high or low values that have become part of a data set as a result of data entry errors?
   (a) A function that averages numeric values in a column or row
   (b) A function that counts the values in a column or row
   (c) A function that rounds a numeric value
   (d) A function that sorts values in a column or row
4. What is the output of the statement "a" + "ab" in Python 3?
   (a) Error
   (b) aab
   (c) ab
   (d) a ab
5. What is the output of 4*1**4 in python?
   (a) 4
   (b) 1
   (c) 256
   (d) 16

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, the expression x >> 1 results in the value 4 when x is equal to 8.

7. True or False: In Python 3, the len() function checks if all characters in a string are uppercase.

8. True or False: In Python 3, the random() function sets the integer starting value used in generating random numbers.

9. True or False: The condition a < b is sufficient to ensure that the expression a < c || a < b && !(a == c) is true.

10. True or False: If no errors are found during testing, it guarantees that all methods can be safely used in other programs.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Explain how the RGB color model represents colors using red, green, and blue values. Discuss the total number of bits required for this representation and the significance of this choice.

12. Discuss the operator in Python 3 that is used for performing exponential calculations on operands. Explain its syntax, usage, and provide examples to illustrate its functionality.

End of Paper